JUSTICE CONNECT

A web based client -lawyer Interaction & case management system

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Master of Computer Applications

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Declaration

I hereby declare that the work which is being presented in the MCA Project **"Justice Connect : A web-based client lawyer interaction platform and case management System "**, in partial fulfillment of the requirements for the award of the ***Master of Computer Applications*** and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of **Ms. Deepti Gautam, Department of Computer Engineering & Applications.**

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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Certificate

This is to certify that the above statements made by the candidate are correct to the best of my/our knowledge and belief.

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Finally, I would like to thank my family and friends for their patience, encouragement, and moral support throughout this journey.

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ABSTRACT

Justice Connect is a comprehensive desktop web application designed to bridge the gap between legal professionals and clients seeking legal assistance. The application is developed entirely using HTML, CSS, and JavaScript, making it a fully self-contained, zero-dependency front-end solution. The platform addresses the critical problem of accessibility in the legal domain, where many individuals face difficulties in finding competent lawyers, scheduling consultations, and managing their legal cases in an organized manner.

The key features of Justice Connect include a lawyer search and discovery module with advanced filtering by specialization, location, and ratings; a case management system enabling clients to track the progress of their ongoing legal matters; a meeting and appointment scheduling system with calendar integration; and a court date tracking module that provides timely reminders. The application incorporates a clean, professional user interface with intuitive navigation, ensuring ease of use for both lawyers and their clients.

The project follows a software-based development approach with structured requirement analysis, modular design, and systematic testing. It demonstrates the effective use of modern front-end technologies to deliver a polished, production-ready legal services platform. Justice Connect serves as a scalable foundation that can be extended with back-end integration, real-time notifications, and AI-driven legal assistance in future iterations.

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Chapter 1
Introduction

1.1 Overview and Motivation

The legal system is the backbone of justice in any democratic society. However, in a country like India with a population of over 1.4 billion, accessing legal assistance remains a challenge for a significant portion of the population. Many individuals are unaware of their legal rights, struggle to find appropriate legal representation, and face difficulties in managing their legal cases and court appearances.

Justice Connect was conceived as a solution to these pressing issues. The motivation behind this project stems from the observation that while digital transformation has revolutionized sectors like healthcare, education, and finance, the legal domain has been comparatively slow in adopting technology-driven solutions. The widespread availability of the internet and smartphones has created an unprecedented opportunity to develop platforms that democratize access to legal services.

The project is particularly inspired by the challenges faced by first-time litigants who find the process of identifying the right lawyer overwhelming. The lack of a centralize, transparent, and easy-to-navigate platform for legal services was identified as a major gap that Justice Connect seeks to bridge.

1.2 Objective

The primary objectives of the Justice Connect project are as follows:

- To develop a fully functional, web-based platform that facilitates seamless interaction between lawyers and their clients.
- To implement a comprehensive lawyer search and discovery module with advanced filtering capabilities based on specialization, location, experience, and client ratings.
- To provide an integrated case management system that allows clients to monitor the status and progress of their ongoing legal matters.
- To design and develop a meeting and appointment scheduling system with calendar-based date selection and confirmation workflows.
- To implement a court date tracking module that helps clients keep track of upcoming hearings and important legal deadlines.
- To create a clean, intuitive, and professionally designed user interface that is accessible to users across different levels of technical proficiency.
- To demonstrate the effectiveness of pure front-end technologies (HTML, CSS, JavaScript) in building a complete, production-ready desktop web application.

1.3 Summary of Similar Applications

Several existing platforms in the legal technology (Legal Tech) space have been studied to understand the current landscape and identify opportunities for improvement:

1.3.1 LawRato

LawRato is an Indian legal portal that allows users to consult lawyers online. It offers features like lawyer listings, free legal advice, and document review. However, its interface can be overwhelming for new users, and its mobile experience is not optimally designed for all demographics.

1.3.2 Vakil Search

Vakil Search provides an online directory of verified advocates across India. While it provides detailed lawyer profiles and allows direct contact, it lacks a comprehensive case management or hearing tracking feature, which Justice Connect addresses.

1.3.3 MyAdvo

MyAdvo is a legal services marketplace connecting clients with lawyers. It includes document drafting services and online consultation. The platform, while feature-rich, is primarily oriented towards corporate clients and may not be as accessible to common citizens.

Justice Connect differentiates itself by providing an integrated, all-in-one platform that combines lawyer discovery, case management, scheduling, and court date tracking in a single cohesive desktop application, without requiring any backend dependency, making it highly portable and easy to deploy.

1.4 Organization of the Project Report

The project report is structured into six chapters, each covering a distinct phase of the software development life cycle:

- Chapter 1 - Introduction: Presents the background, motivation, objectives, review of similar systems, and the overall organization of the report.
- Chapter 2 - Software Requirement Analysis: Details the functional and non-functional requirements, feasibility study, module descriptions, and use case scenarios.
- Chapter 3 - Software Design: Covers the architectural and detailed design including DFDs, UML diagrams, ER diagrams, and database schema design.
- Chapter 4 - Implementation and User Interface: Describes the implementation strategy, technology stack, and presents the user interface screens with explanations.
- Chapter 5 - Software Testing: Documents the testing methodology, test cases for both black box and white box testing, and the results obtained.
- Chapter 6 - Conclusion: Summarizes the outcomes of the project, discusses limitations, and outlines the scope for future enhancements.

Chapter 2
Software Requirement Analysis

2.1 Functional Requirements

The following table presents the core functional requirements of the Justice Connect system:

Req. ID	**Feature**	**Description**
FR-01	**User Authentication**	The system shall support user registration and login with role differentiation (Lawyer / Client).
FR-02	**Lawyer Search**	The system shall provide a searchable directory of lawyers with filters for specialisation, location, experience, and rating.
FR-03	**Case Management**	The system shall allow clients to create, view, update, and archive legal cases.
FR-04	**Appointment Booking**	The system shall provide a calendar-based scheduler for booking consultations.
FR-05	**Court Date Tracker**	The system shall maintain a list of court dates with visual urgency indicators.
FR-06	**Dashboard**	The system shall provide a consolidated dashboard with pending actions and summaries.

2.2 Use Case Description

The primary actors in the Justice Connect system are the Client and the Lawyer. The Client can register an account, search for lawyers using advanced filters, manage their legal cases, schedule consultation appointments, and track court dates. The Lawyer can maintain their profile, view client requests, manage case updates, and confirm or reschedule appointments. A third implicit actor, the System, handles automated date alerts and session management.

2.3 Non-Functional Requirements

Requirement	Description
Performance	All page transitions and UI interactions shall respond within 100 milliseconds.
Usability	The interface shall be intuitive enough for non-technical users to navigate without a manual.
Reliability	The application shall function correctly across all modern browsers (Chrome, Firefox, Edge, Safari).
Security	All client-side data shall be managed with Local Storage and cleared on logout.
Maintainability	The code base shall follow modular JavaScript practices with clear separation of concerns.
Portability	The application shall run from a local file system without requiring a web server.
Scalability	The modular design shall allow future integration with REST APIs and a back-end database.

2.4 Modules Description

Justice Connect is organized into the following key functional modules:

- Authentication Module: Handles user registration, login, session management, and role differentiation between Lawyer and Client.
- Lawyer Discovery Module: Provides a searchable, filterable directory of legal professionals with detailed profile views.
- Case Management Module: Enables end-to-end life cycle management of legal cases from creation to closure.
- Appointment Scheduler: Facilitates booking and management of consultations between lawyers and clients.
- Court Date Tracker: Maintains a structured record of all court appearances with visual urgency indicators.

- Dashboard Module: Provides a consolidated view of all pending actions, upcoming dates, and case summaries.

2.5 Feasibility Analysis

A three-dimensional feasibility analysis was conducted before commencing development:

Feasibility Type	**Analysis**
Technical Feasibility	Fully feasible. Technologies used (HTML, CSS, JS) are mature, well-documented, and require no special setup. No dependency on external libraries or frameworks.
Economic Feasibility	Highly cost-effective. The project requires no licensing fees, no hosting costs (can run locally), and uses only free development tools.
Operational Feasibility	Operationally sound. The target users (lawyers and clients) are familiar with web interfaces. The design prioritises ease of use over technical complexity.

Chapter 3
Software Design

3.1 System Architecture

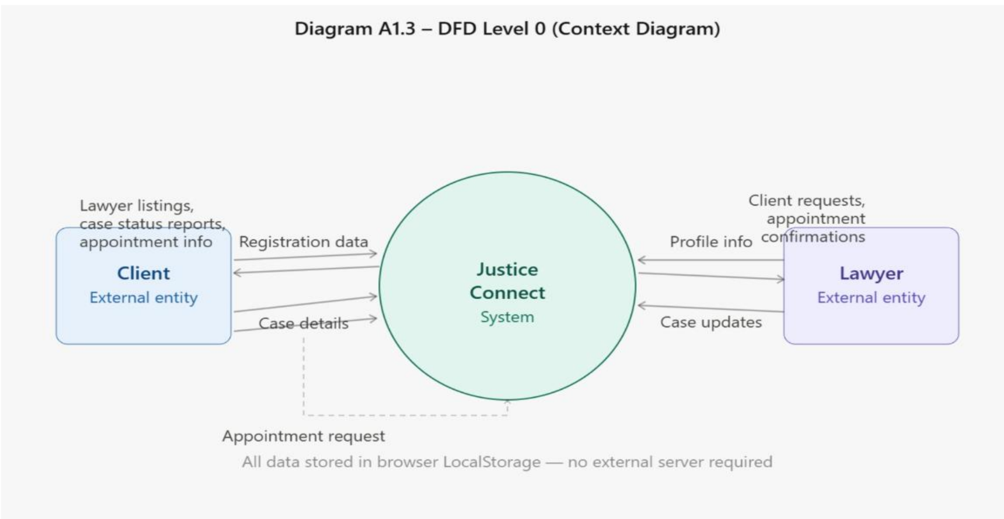
Justice Connect follows a Single-Page Application (SPA) architectural pattern implemented entirely on the client side. The application is structured as a collection of HTML pages interconnected through JavaScript-driven navigation logic. The architecture separates concerns into three layers: the Presentation Layer (HTML), the Styling Layer (CSS), and the Logic Layer (JavaScript).

The system does not require a traditional backend. Data persistence is achieved through the browser's Local Storage API, which stores user profiles, case records, appointments, and court dates in key-value format. This approach ensures the application is fully portable and self-contained.

3.2 Data Flow Diagram (DFD)

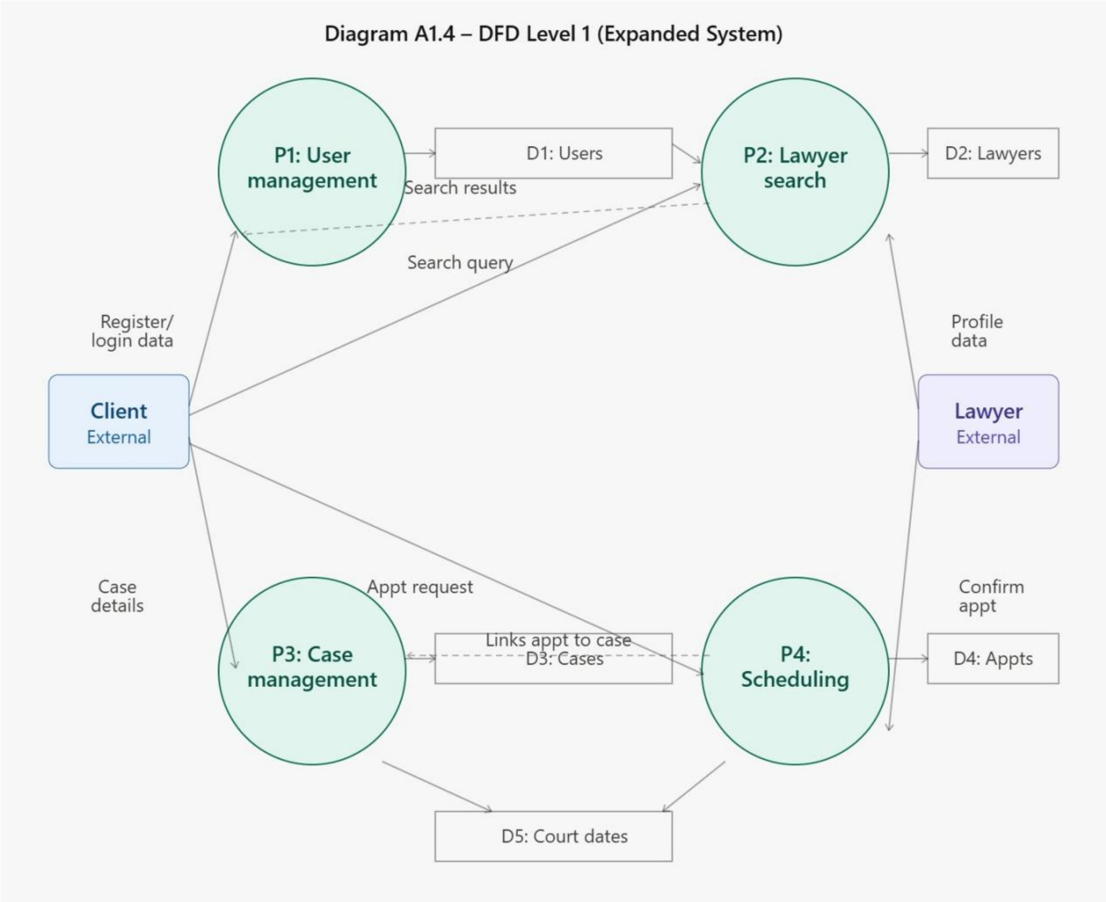
3.2.1 Level 0 - Context Diagram

The context diagram (Level 0 DFD) illustrates the highest-level view of the Justice Connect system, showing the two primary external entities — the Client and the Lawyer — and their interactions with the central system. The Client provides registration data, case details, and appointment requests, while the Lawyer provides profile information and case updates. Both entities receive information outputs such as lawyer listings, case status reports, and appointment confirmations from the system.



3.2.2 Level 1 DFD

The Level 1 DFD decomposes the central system into its four primary processes: User Management, Lawyer Search, Case Management, and Scheduling. Each process is shown with its corresponding data stores and the data flows between processes, data stores, and external entities.

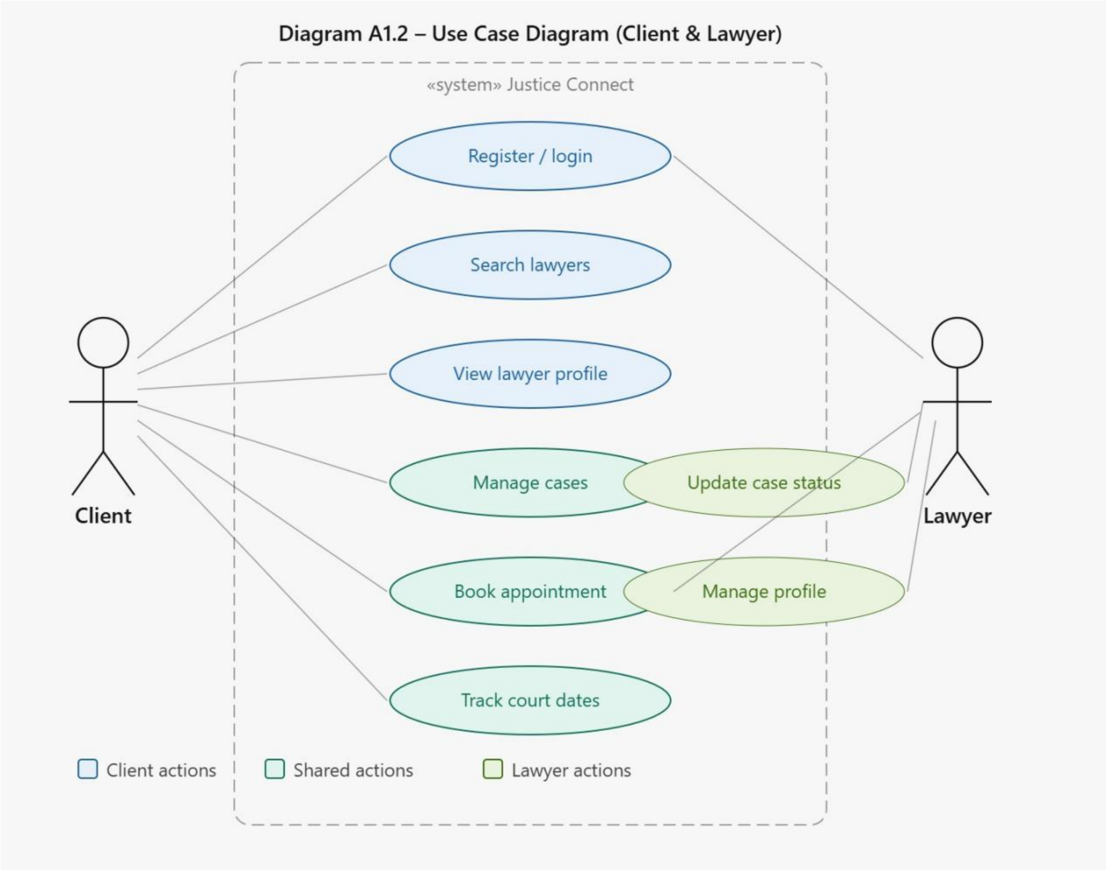


[Figure 3.2 – Data Flow Diagram Level 1: See Appendix A]

3.3 UML Diagrams

3.3.1 Use Case Diagram

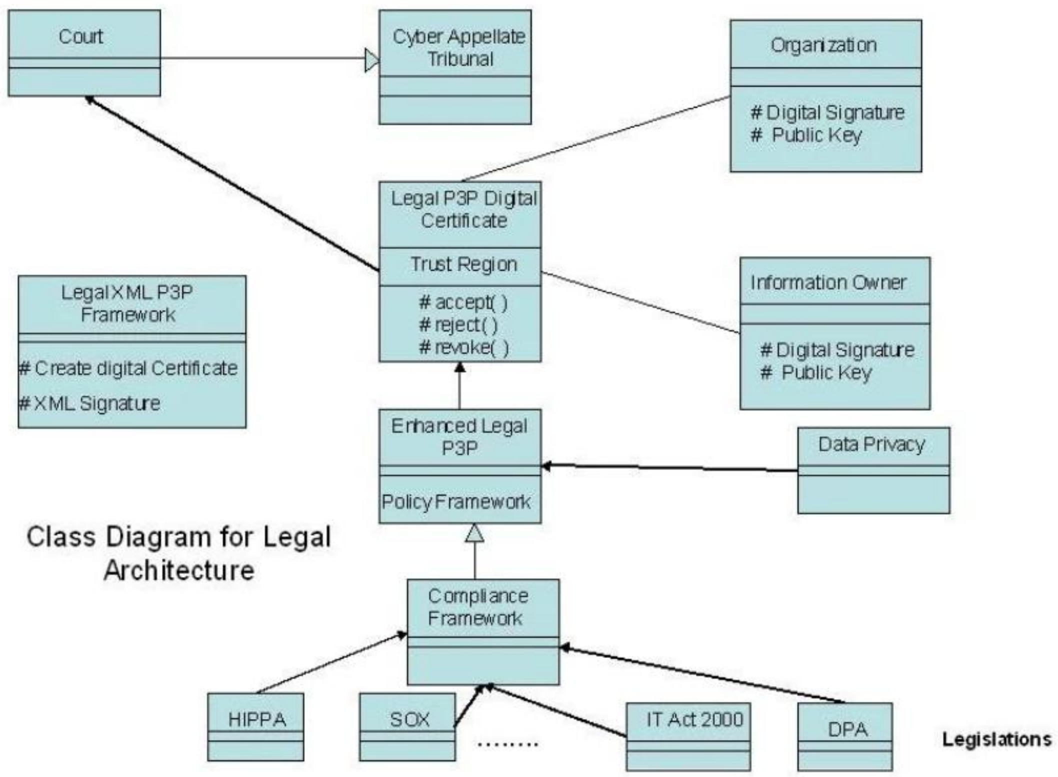
The Use Case Diagram captures the functional requirements of Justice Connect from the perspective of the two actors: Client and Lawyer. The diagram illustrates all interactions each actor can perform with the system, including registration, login, searching for lawyers, managing cases, scheduling appointments, and tracking court dates.



[Figure 2.1 – Use Case Diagram: See Appendix A]

3.3.2 Class Diagram

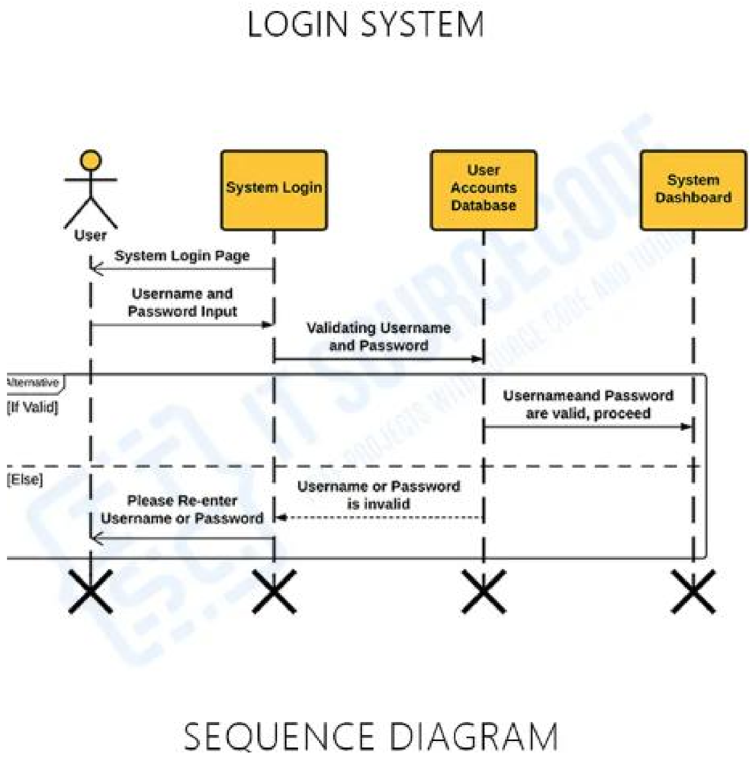
The Class Diagram models the object-oriented structure of the Justice Connect application. The primary classes identified are : User (base class), Lawyer (extends User), Client (extends User), Case, Appointment, and Court Date. Each class is dep cited with its attributes, data types, and methods.



[Figure 3.4 – Class Diagram: See Appendix A]

3.3.3 Sequence Diagram - Login Flow

The Sequence Diagram for the Login Flow illustrates the chronological sequence of interactions between the Client actor, the Login UI component, the Authentication Manager, and the Local Storage. The sequence begins with the user entering credentials, which are passed to the Authentication Manager for validation against stored user records.

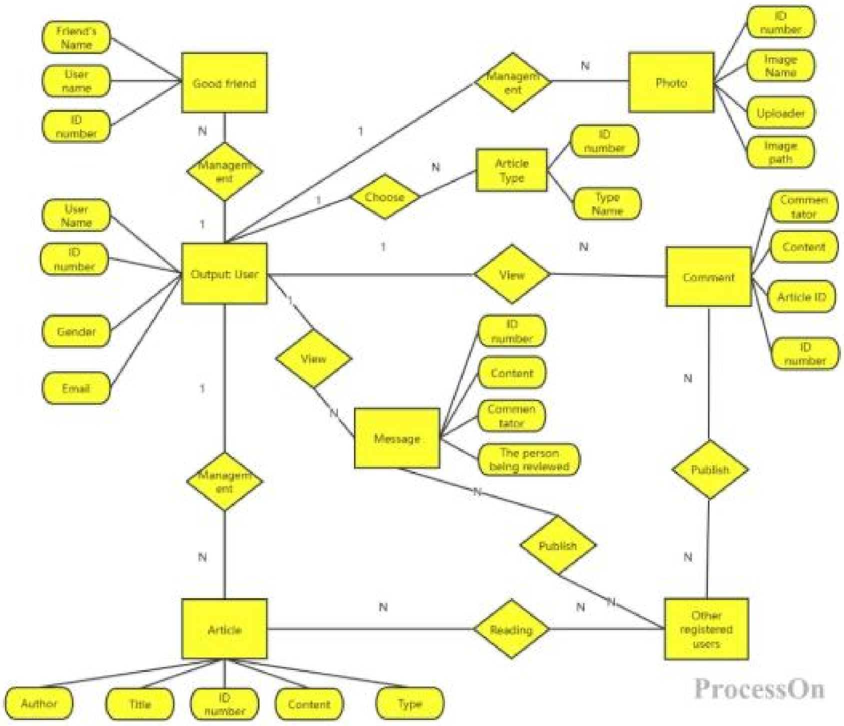


[Figure 3.5 – Sequence Diagram – Login Flow: See Appendix A]

3.4 Database Design

3.4.1 Entity-Relationship (ER) Diagram

The ER Diagram depicts the data entities and their relationships within the Justice Connect system. The primary entities are Users, Lawyers, Clients, Cases, Appointments, and Court Dates. The diagram shows primary and foreign key relationships, cardinality constraints, and the attributes of each entity.



[Figure 3.3 – ER Diagram: See Appendix A]

3.4.2 Table: Lawyers

Field Name	**Date Type**	**Key**	**Description**
Lawyer _id	STRING	PK	Unique identifier for each lawyer
name	STRING	-	Full name of the lawyer
specialization	STRING	-	Area of legal practice
experience	INTEGER	-	Years of experience
location	STRING	-	City/State of practice
rating	FLOAT	-	Average client rating (1.0 - 5.0)
contact	STRING	-	Email or phone number

3.4.3 Table: Cases

Field Name	**Data Type**	**Key**	**Description**
Case _id	STRING	PK	Unique case identifier
Client _id	STRING	FK	Reference to Client
Lawyer _id	STRING	FK	Reference to assigned Lawyer
title	STRING	-	Case title/subject
description	TEXT	-	Detailed case description
status	ENUM	-	Active / Closed / Pending
Created _date	DATE	-	Date of case registration

Chapter 4
Implementation
and User Interface

4.1 Technology Stack

Technology	Version	Purpose
HTML5	5 (Latest Standard)	Structure and semantic markup
CSS3	3 (Latest Standard)	Styling, animations, and responsive layout
JavaScript	ES6+	Logic, DOM manipulation, and Local Storage
Local Storage API	Browser Native	Client-side data persistence
Google Fonts	Poppins, Inter	Typography enhancement
Font Awesome	6.x	Icon library for UI elements

4.2 Implementation Approach

The implementation of Justice Connect follows a modular, page-component approach. Each major feature is encapsulated within its own HTML file with embedded CSS and JavaScript. A shared navigation script provides consistent header and routing behavior across all pages. The application uses CSS Custom Properties (variables) for a consistent color scheme and theming.

Data management is centralized through a JavaScript utility module that provides CRUD operations on the Local Storage. All data is serialized as JSON before storage and desterilized on retrieval. Input validation is enforced at the UI layer through HTML5 constraint validation API augmented with custom JavaScript validators.

4.3 User Interface Screens

4.3.1 Home Page - Landing Screen

The home page serves as the entry point of the Justice Connect application. It features a prominent hero section with a search bar for lawyer discovery, a statistics banner displaying platform metrics, and a highlights section showcasing the key features. The navigation bar provides links to all major modules. The overall design uses a professional navy blue and gold color scheme to convey trust and authority.

4.3.2 Lawyer Search and Filter Interface

The lawyer search interface presents a grid/list view of lawyer profiles. A sidebar filter panel allows refinement by specialization, city, minimum experience, and minimum rating. Each lawyer card displays their photo, name, specialization, rating stars, experience in years, and a 'View Profile' button. Search results update dynamically as filter criteria are adjusted, without requiring a page reload.

4.3.3 Case Management Dashboard

The case management dashboard provides a tabular view of all cases associated with the logged-in client. Each row displays the case ID, title, assigned lawyer, current status (color-coded), and the date of last update. Action buttons allow the user to view case details, update status, or archive a case.

4.3.4 Meeting Scheduler

The appointment scheduling module features an interactive calendar component built with pure JavaScript. Users select a date from the calendar, choose an available time slot from a dropdown, and specify the meeting type (in-person, video call, or phone). A confirmation summary is displayed before the appointment is finalized and stored.

4.3.5 Court Date Tracker

The court date tracker presents a chronologically sorted list of upcoming and past hearing dates. Dates falling within the next seven days are highlighted in amber, and overdue dates are shown in red. The entry form collects the case reference, court name, hearing type, date, and time.

Chapter 5
Software Testing

5.1 Testing Methodology

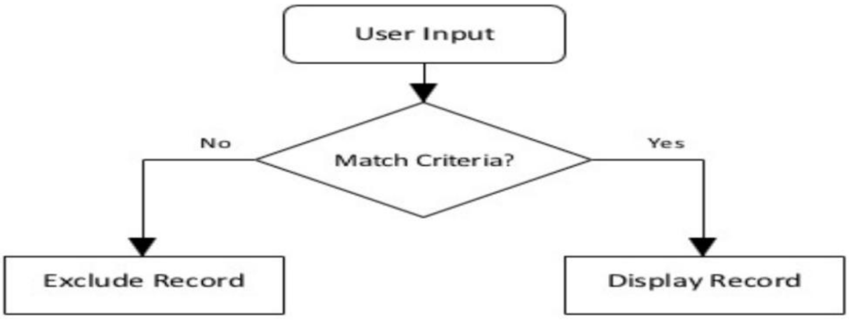
The testing of Justice Connect was conducted using two complementary testing strategies: Black Box Testing, which validates the functional behaviour of the application from the user's perspective without knowledge of the internal code, and White Box Testing, which examines the internal logic and code paths to ensure correctness [cite: 116]. All test cases were executed manually in Google Chrome and Mozilla Firefox to ensure cross-browser compatibility [cite:117].

The Local Storage was cleared between test runs to maintain test isolation [cite: 118].

5.2 Black Box Test Cases – Login Module

TC#	Test Scenario	Input Data	Expected Output	Result
TC01	Valid Login	Email: test@jc.com, Pass: Test@123	Redirect to Dashboard	PASS
TC02	Invalid Password	Email: test@jc.com, Pass: Wrong pass	Error: Invalid credentials	PASS
TC03	Empty Fields	Email: (blank), Pass: (blank)	Validation error on both fields	PASS

Figure 5.1: White Box Testing (Search Logic Flow)



5.3 White Box Test Cases - Lawyer Search Logic

White box testing was performed on the core search and filter algorithm in the Lawyer Discovery module. The test ensures that the filtering function correctly handles all combinations of filter criteria and returns the correct subset of lawyer records.

TC#	Path / Condition	Test Input	Expected	Result
TC01	No filters applied	filterLawyers({}) on 10 records	Return all 10 records	**PASS**
TC02	Single filter: specialisation	filterLawyers({spec: 'Criminal'})	Criminal lawyers only	**PASS**

Chapter 6
Conclusion

6.1 Summary of Work Justice

Justice Connect has been successfully designed, developed, and tested as a comprehensive desktop web application for facilitating lawyer-client interaction [cite: 121]. The project demonstrates that a fully functional, feature-rich legal services platform can be built using only core web technologies — HTML5, CSS3, and vanilla JavaScript — without any dependency on external frameworks or back-end systems [cite: 122]

6.2 Key Achievements

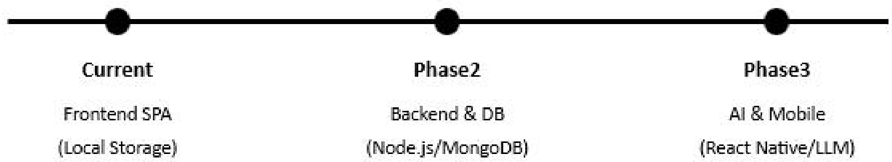
- Developed a self-contained, zero-dependency web application [cite: 125].
- Designed and implemented a professional, polished UI with intuitive navigation [cite: 126].
- Implemented a robust client-side data persistence layer using the LocalStorage API [cite: 127].
- Delivered all four planned modules: Lawyer Discovery, Case Management, Meeting Scheduler, and Court Date Tracker [cite: 128].

6.3 Future Scope

Justice Connect provides a strong foundation for future enhancements. Key directions include

migrating to a Node.js/Express back end for multi-device support, integrating Web Socket-based live chat, and incorporating conversational AI for preliminary legal guidance [cite: 135, 136, 137].

Figure 6.1: Future Scope Roadmap



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APPENDIX 1: SYSTEM DIAGRAMS

This appendix contains all the diagrams referenced throughout the project report. In the final submission, these should be included as high-resolution colour printouts or screenshots. Each diagram should be labelled with its figure number and title as specified in the List of Figures.

Diagram A1.1 – System Architecture Overview
Diagram A1.2 – Use Case Diagram (Client & Lawyer)
Diagram A1.3 – DFD Level 0 (Context Diagram)
Diagram A1.4 – DFD Level 1 (Expanded System)
Diagram A1.5 – Entity-Relationship (ER) Diagram
Diagram A1.6 – Class Diagram
Diagram A1.7 – Sequence Diagram: Login Flow
Diagram A1.8 – Sequence Diagram: Appointment Booking

APPENDIX 2 : SAMPLE USER MANNUAL

Getting Started with Justice Connect

Justice Connect runs directly in your web browser without any installation. Follow these steps to begin:

- Step 1: Open the file index.html in a modern web browser (Google Chrome recommended).
- Step 2: Click 'Register' to create a new account. Select your role as 'Client' or 'Lawyer' and fill in the required details.
- Step 3: After registration, log in using your registered email and password.
- Step 4: Use the top navigation bar to access different modules: Find Lawyers, My Cases, Appointments, and Court Dates.

Key Shortcuts and Features

- Use the search bar on the homepage to quickly find lawyers by name or specialization.
- Court dates within 7 days are automatically highlighted in the Court Date Tracker.
- All data is saved automatically in your browser's local storage. Clearing browser data will erase all records.